

Mohamed Mamdouh

Full-Stack Developer | .NET | Angular

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SUMMARY

Full-Stack .NET & Angular Developer focused on building scalable web applications with ASP.NET Core, Clean Architecture, and Angular. Skilled in writing clean, maintainable, and testable code.

EDUCATION

Bachelor of Engineering in Computer and Systems Engineering,
Zagazig University — 2026.

TECHNICAL SKILLS

Languages: C#, C++, TypeScript, JavaScript, SQL, T-SQL, HTML, CSS.

Backend Frameworks: ASP.NET Core Web API, Entity Framework Core, ADO.NET, ASP.NET Identity, JWT & httpOnly Refresh Tokens, Role-Based Authorization (RBAC), Dependency Injection, FluentValidation, MediatR (CQRS), SignalR, Background Services, Caching, Middlewares, Observability.

Frontend Frameworks: Angular, RxJS, Lazy Loading, Angular Routing, Reactive Forms, HTTP Interceptors, Route Guards, Services & Dependency Injection, Component Communication, Angular Signals, State Management, Responsive Design, RTL Support, Tailwind CSS, Bootstrap.

Databases & Storage: SQL Server, PostgreSQL, MongoDB, Cloudflare R2, Database Design, Query Optimization.

Testing & Deployment: xUnit, Testcontainers, Docker, GitHub Actions (CI/CD), Postman, Swagger, Git, GitHub.

Architecture & Concepts: SOLID Principles, Design Patterns, Clean Code, Clean Architecture, Domain-Driven Design (DDD) fundamentals, Transactional Outbox Pattern, RESTful API Design, SDLC, OOP, Algorithms, Data Structure.

SELECTED PROJECTS

Amanah – Lost & Found Platform

Full-stack production lost-and-found platform, handling report submissions, moderation queues, ownership claims, and real-time chat. [GitHub](#) [Live Production](#)

Tech Stack: .NET, Angular, PostgreSQL, EF Core, FluentValidation, JWT Bearer, SignalR, Cloudflare R2, Docker, xUnit, Testcontainers, GitHub Actions.

Key Highlights:

- Engineered API reliability with correlation IDs, centralized exception handling, field-level validation, configurable rate limiting, HybridCache, structured logging, and PostgreSQL/R2 readiness checks.
- Implemented secure authentication workflows featuring phone/email OTP registration, password authentication, short-lived JWTs, and httpOnly refresh cookies for session persistence.
- Developed responsive Angular interfaces with native RTL/Arabic support, reactive form validations, EXIF metadata stripping, and client-side WebP image processing prior to Cloudflare R2 upload.
- Integrated bidirectional real-time chat via SignalR with REST fallback policies, coupled with a Transactional Outbox pattern to guarantee reliable SMS and email dispatch.
- Implemented background workers for listing expiration, claim timeouts, orphaned storage cleanup, and scheduled maintenance tasks.
- Built integration tests with xUnit and Testcontainers using isolated database instances, and integrated them into GitHub Actions CI pipelines.

Shora – Full-Stack Consulting & Booking Platform

[GitHub](#) [Live Production](#)

Full-stack production booking platform for one-to-one consulting sessions with availability management, manual payment verification, client bookings, and admin operations.

Tech Stack: .NET, Angular, PostgreSQL, EF Core, ASP.NET Core Identity, JWT Bearer, Cloudflare R2, Docker, xUnit, Testcontainers, GitHub Actions.

Key Highlights

- Clean Architecture: .NET API with EF Core/PostgreSQL and Angular standalone SPA, deployed as a single Docker image.
- Authentication & session security: JWT access tokens, rotating httpOnly refresh tokens, row locking, reuse detection, and secure session restoration.
- Booking & payments: Implemented slot holds, concurrency control, expiry jobs, and manual payment workflows with receipt verification and refunds.
- Reliability & integrations: Built a transactional outbox for emails, private Cloudflare R2 storage, background workers, rate limiting, caching, and Problem Details.
- Testing & CI: Built xUnit integration tests with Testcontainers PostgreSQL and automated CI workflows.

MechanicShop – Workshop Operations API

[GitHub](#)

Backend REST API for a mechanic shop covering work orders, day scheduling, labor assignment, customer and vehicle records, and invoicing.

Tech Stack: ASP.NET Core, SQL Server, EF Core, MediatR, JWT, ASP.NET Identity, FluentValidation, Docker, xUnit, Testcontainers, GitHub Actions.

Key Highlights

- Architected a Clean Architecture backend with CQRS/MediatR, Result Pattern, FluentValidation, and pipeline behaviors for validation, logging, transactions, and caching.
- Modeled shop operations with rich aggregates for work orders, scheduling, labor assignment, repair tasks, and invoicing.
- Implemented JWT authentication with rotating refresh tokens, per-device sessions, RBAC, and custom authorization for assigned mechanics.
- Hardened the API with optimistic concurrency, Problem Details, output caching, rate limiting, and SQL Server application locks.
- Built unit and integration tests with Testcontainers and automated CI with separate GitHub Actions jobs.

DVLD – Driver & Vehicle License Department System

[GitHub](#)

Desktop application for managing driver licenses, license applications, driving tests, and vehicle-related records.

Tech Stack: C#, .NET Framework, Windows Forms, ADO.NET, SQL Server.

Key Highlights:

- Developed a layered WinForms solution in .NET Framework (UI, business, data access) to keep presentation, domain behavior, and persistence decoupled.
- Connected to SQL Server using ADO.NET and parameterized stored procedures for entity CRUD, reference lookups, and rule-driven validation queries.
- Applied licensing domain modeling and rules administration with consistent enforcement at application and exam stages.
- Completed full operational flows, license transactions, vision/written/street testing with retakes, and authenticated user access via password hashing.